

Morning Paper Report - Tuesday, September 22, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Implicit learning in visual search: the role of spatial and object working memory

K. V. Signaevskaya, E. Gorbunova

Sibirskiy psikhologicheskiy zhurnal, 2026

Source: Semantic Scholar; matched terms: visual working memory, visual search, working memory

Link: <https://www.semanticscholar.org/paper/5243a96f67f0fac1efef2c20453a481562db5f73>

Goal: The contextual cueing effect, consisting in a gradual reduction of target search time with repeated context, is traditionally considered a form of implicit learning; however, its dependence on working memory resources remains a subject of active discussion

Method: The aim of this work is to investigate the role of spatial and object working memory in the emergence and expression of the contextual cueing effect, considering individual differences in working memory capacity among participants

Results: The results of studies on the influence of working memory load on this effect are ambiguous, which is due to the lack of consideration for the multicomponent structure of visual working memory and individual differences in its capacity, creating a prerequisite for the present study

Conclusion: The data indicate a functional dissociation of visual working memory into components and emphasize the role of memory capacity in the observed effect

2. Repulsive serial dependence in visual working memory modulated by target fidelity

Bugay Yildirim, Aysecan Boduroglu

BMC Biology, 2026

Source: Crossref; matched terms: visual working memory, working memory, fidelity

Link: <https://doi.org/10.1186/s12915-026-02739-9>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

3. Effects of working memory and visual search skill on Chinese reading comprehension: examining the simple view of reading

Sisi Liu, Ning Li, Xinyong Zhang, Li-Chih Wang, D. Liu

Reading & Writing, 2024

Source: Semantic Scholar; matched terms: visual search, working memory, modeling

Link: <https://www.semanticscholar.org/paper/6c681a40e8d4ce8053aaa29adb7268d19465f057>

Goal: To investigate the longitudinal effects of two domain-general cognitive abilities, namely verbal working memory and visual search skill, on Chinese reading comprehension

Method: A total of 202 first-grade Chinese-speaking children from mainland China (Mage = 86 months, SD = 4.9 months; 126 boys) completed tasks that measured verbal working memory, visual search, expressive vocabulary, morphological awareness, Chinese character recognition, and word reading fluency

Results: In line with the simple view of reading theory, decoding and linguistic comprehension skills explain such impacts This research underlines the value of simultaneously examining general cognitive abilities and exploring their interconnectedness with linguistic skills in the context of reading comprehension.

Conclusion: Our findings suggest that the reading comprehension ability of beginning Chinese readers is associated with verbal working memory and visual search ability

4. Human Beta Oscillations Reflect Magnitude and Fidelity of Priority Shifts in Working Memory

Nicholas E. Myers, Mark G. Stokes, Paul S. Muhle-Karbe
2025

Source: Crossref; matched terms: working memory, fidelity

Link: <https://doi.org/10.1101/2025.08.07.669112>

Goal: Abstract Flexible prioritisation in working memory (WM) is supported by neural oscillations in frontal and sensory brain areas, but the roles of different oscillations remain poorly understood

Method: Here we investigated the role of neural oscillations in prioritising WM content using electroencephalography (EEG) as participants (humans of any sex) performed a task requiring frequent priority switches between two memorized oriented bars

Results: Behavioural performance revealed switch costs, which scaled with the angular distance between the two items, suggesting that priority shifts are modulated by shift magnitude

Conclusion: Abstract Flexible prioritisation in working memory (WM) is supported by neural oscillations in frontal and sensory brain areas, but the roles of different oscillations remain poorly understood

5. Differential Contributions of Inhibitory Subnetwork to Visual Cortical Modulations Identified via Computational Model of Working Memory

William H. Nesse, Zahra Bahmani, Kelsey Clark, Behrad Noudoost
Frontiers in Computational Neuroscience, 2021

Source: Crossref; matched terms: working memory, computational model

Link: <https://doi.org/10.3389/fncom.2021.632730>

Goal: Extrastriate visual neurons show no firing rate change during a working memory (WM) task in the absence of sensory input, but both $\alpha\beta$ oscillations and spike phase locking are enhanced, as is the gain of sensory responses

Method: This lack of change in firing rate is at odds with many models of WM, or attentional modulation of sensory networks

Results: Extrastriate visual neurons show no firing rate change during a working memory (WM) task in the absence of sensory input, but both $\alpha\beta$ oscillations and spike phase locking are enhanced, as is the gain of sensory responses

Conclusion: Using a phase oscillator model we were able to show that inhibitory tone is principally modulated through inhibitory network gain saturation, while the phase-dependent efficacy of inhibitory currents drives the phase locking modulation The dual contributions of the inhibitory subnetwork to oscillatory and non-oscillatory modulations of neural activity provides two distinct ways for WM to recruit sensory areas,...